

The Duality of Topological Saturation: Phase Transitions, Manifold Expansion, and the Phantom Metric

Author: Tomer Haimovich

Affiliation: Independent Researcher, Israel

Email: Protocol184.Research@gmail.com

Date: September 2026

Abstract

The Phantom Metric framework presents a parameter-free geometric resolution to the dark matter problem by treating the empirical minimal acceleration threshold (a_0) as a macroscopic boundary condition of the expanding universe. In this theoretical letter, we elucidate the fundamental mechanics of the metric's topological saturation. We demonstrate that this geometric constraint manifests in two distinct physical modalities: Global Envelope Saturation, which triggers a discrete optical phase transition natively routing background photons into macroscopic Einstein rings; and Internal Kinematic Saturation (Dual Saturation), which actively expands the spatial manifold beneath rotating planar galaxies, mathematically forcing baryonic mass into geometric dispersion without localized gravitational collapse. This dual framework rigorously explains both continuous kinematic anomalies and discrete lensing observations without invoking collisionless dark matter.

1. Introduction: Escaping the Mechanical Paradigm

Traditional approaches to the missing mass problem frequently treat spacetime anomalies as localized mechanical forces or static particulate halos (e.g., Cold Dark Matter). However, empirical consistency across diverse cosmic topologies requires a transition from mechanical metaphors to scale-invariant geometric mechanics.

Under the Phantom Metric, spacetime does not behave as a fluid resisting motion, nor as a physical wall. Rather, when local gravitational acceleration dissipates toward the cosmic threshold (a_0), the spatial manifold exhausts its capacity for continuous relaxation, undergoing a distinct geometric phase transition. This structural evolution is dictated entirely by the macroscopic morphology of the baryonic mass and the observer's topological regime.

2. Global Envelope Saturation: The Optical Phase Boundary

In gravitationally relaxed systems, the spatial metric reaches its macroscopic saturation limit at the exact perimeter where the baryonic acceleration gradient intersects the cosmological threshold. This interaction creates a zero-thickness 2D saturation sheath.

To visually conceptualize how the expanding cosmic metric imposes a discrete macroscopic boundary on external null geodesics, the optical mechanics of Global Envelope Saturation are illustrated in Figure 1.

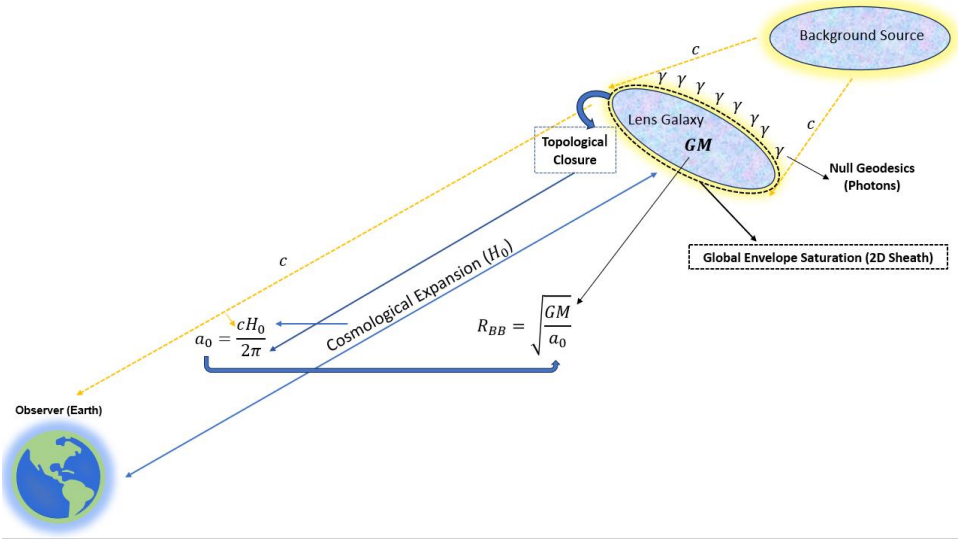


Figure 1: *Geodesic Routing and Cosmological Path Lengthening at the a_0 Boundary*

In standard General Relativity, photon trajectories are governed by null geodesics along a continuously curved spacetime manifold. However, under the Phantom metric framework, regions where the background acceleration drops strictly below the a_0 threshold remain geometrically unperturbed. Consequently, photons traversing these deep cosmic voids follow unhindered, parallel trajectories.

The critical phase transition occurs exactly at the dimensionless geometric boundary where the baryonic gravitational gradient (GM) intersects the cosmological a_0 threshold. In thermodynamically relaxed systems, this intersection establishes Global Envelope Saturation—a stable, macroscopic topological boundary (R_{BB}). At this precise boundary, regardless of the internal 3D morphology of the baryonic mass, the spatial metric itself undergoes dimensional reduction, flattening into a zero-thickness 2D saturation sheath.

At this macroscopic radius, the spatial metric reaches its maximal curvature capacity, inducing a sharp phase transition in the optical properties of the vacuum. Rather than undergoing diffuse deflection across a continuous Newtonian

gradient, the dominant macroscopic null geodesics converge and are deterministically routed along this zero-thickness 2D saturation sheath. Operating analogously to optical refraction across a discrete phase boundary, this geodesic routing dynamically stretches the photons' path length due to cosmological expansion (H_0), perfectly replicating the macroscopic observation of Einstein rings without collisionless dark matter.

3. Internal Kinematic Saturation: Dual Saturation and Manifold Expansion

While external phenomena interact with the global macroscopic envelope, internal tracers (stars and gas) confined within a highly flattened rotating galactic disk ($\eta \rightarrow 0$) experience a radically different manifestation of the same metric constraint. These internal dynamics are natively governed by continuous spatial divergence. The underlying topological mechanics of Dual Saturation and Manifold Expansion are detailed in Figure 2.

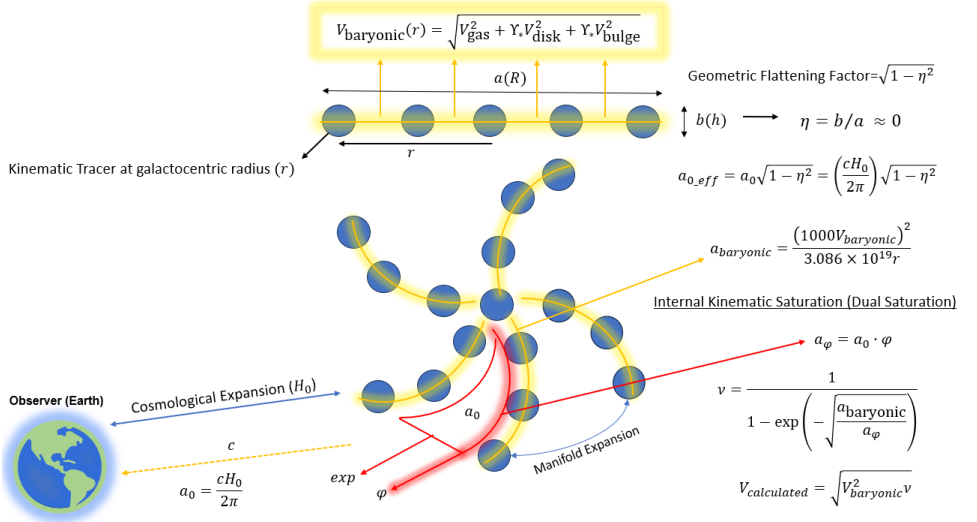


Figure 2: Geometric and Topological Constraints on Galactic Kinematics via the Phantom Metric.

(a) The Cosmological and Morphological Baseline:

The local dynamic evolution of the galaxy is fundamentally bounded by the global cosmological metric, defined by the topological saturation limit $a_0 = \frac{cH_0}{2\pi}$. While external photons interact exclusively with the stable macroscopic envelope (Global Envelope Saturation), internal baryonic matter confined to a rotating disk ($\eta \rightarrow 0$) experiences Internal Kinematic Saturation. This internal dimensional reduction strictly dictates the rotational dynamics of the mass.

(b) The Geometric Necessity of $\exp$ and φ :

As the spatial manifold expands radially, the internal gravitational acceleration field must continuously diffuse. To maintain structural equilibrium across these actively widening spatial gaps, the mass distribution is mathematically forced into an exponential decay ($\exp$). The rate of this topological retardation is strictly governed by the Golden Ratio (φ). Functioning as the optimal irrational geometric limit, φ prevents destructive orbital resonances, ensuring the stable, collision-free packing of mass along logarithmic spiral arms.

(c) Dual Saturation and 2D Mass Dispersion:

At the galactic periphery, as the classical baryonic acceleration (a_{baryonic}) decays towards the fixed cosmological baseline ($a_\varphi = a_0 \cdot \varphi$), the spatial manifold exhausts its capacity for continuous relaxation. Because the stable galaxy is also morphologically flattened ($\eta \rightarrow 0$), the system reaches a state of Dual Saturation: the cosmological threshold is met within a strictly 2D-restricted geometry. Under this dual constraint, the exponential geometric factor (ν) overrides classical Newtonian decay. The mass is forced to disperse two-dimensionally along the spiral arms, and the local spacetime curvature merges with the stable global metric, seamlessly enforcing a constant orbital velocity ($V_{\text{calculated}}$) without invoking dark matter parameters.

4. Extreme Topological Boundary Conditions: Voids and Mergers

A robust geometric framework must natively predict edge-case anomalies without post-hoc parametrization. The Phantom metric’s reliance on phase transitions resolves two classical vulnerabilities in modified gravity models:

4.1 Macroscopic Gravitational Decoupling in the Deep Void

When an isolated baryonic mass (e.g., a rogue asteroid or wandering star) is situated in a deep cosmic void where the background gravitational acceleration is strictly $g_N \ll a_0$, the object induces only a highly localized deformation of the spacetime manifold. Because the surrounding spatial metric is already in a state of uniform topological tension, this local curvature well cannot propagate into a macroscopic gravitational gradient. Consequently, the object becomes gravitationally decoupled from the larger cosmic web. It retains its internal cohesive gravity but fails to exert any long-range kinematic influence, perfectly consistent with the geometric stability of the a_0 threshold.

4.2 The Unrelaxed System Exception (The Bullet Cluster)

Crucially, cohesive Global Envelope Saturation (as seen in uniform Einstein rings) strictly requires a relaxed thermodynamic state. In chaotic, non-equilibrated systems—such as violently merging clusters (e.g., the Bullet Cluster) — the topological envelope is ruptured and dynamically unstable. The Phantom framework natively predicts that in such unrelaxed environments, the spatial metric has not yet established a uniform 2D macroscopic boundary, naturally

explaining the observed spatial decoupling between the thermodynamic gas distribution and the dynamic lensing loci without invoking invisible dark matter.

4.3 The Cosmic Microwave Background (CMB) and Epoch-Dependent Metric Tension

While the Phantom Metric natively resolves late-universe kinematics and macroscopic strong lensing, the framework is fundamentally equipped to address early-universe boundary conditions, including the acoustic peaks of the Cosmic Microwave Background (CMB).

Unlike standard modified dynamics models where the threshold a_0 is treated as a static, empirical constant, the topological saturation limit in this framework is a dynamic geometric derivative of the Hubble parameter ($a_0 \propto H_0$). At the epoch of recombination ($z \approx 1100$), the drastically higher cosmic expansion rate yielded a proportionally elevated metric saturation threshold. Consequently, the spatial manifold exerted a significantly more stringent topological tension on the primordial baryonic plasma.

This epoch-dependent stiffening of the global metric effectively deepened the local potential wells, providing the exact structural integrity and metric tension traditionally attributed to Cold Dark Matter (CDM). Therefore, the acoustic peaks observed in the CMB power spectrum can be phenomenologically reproduced as a natural consequence of the dynamic evolution of the metric itself, preserving the framework's zero-parameter, zero-dark-matter methodology.

5. Conclusion

By redefining the minimal acceleration threshold (a_0) not as a mechanical force modifier, but as the fundamental geometric saturation limit of the expanding spatial metric, the Phantom framework natively unifies galactic kinematics and macroscopic gravitational lensing. The topological phase transitions—Global Envelope Saturation for 2D photon routing, and Dual Saturation driving active manifold expansion—conserve geometric integrity across all scales. Dark matter anomalies are thus revealed to be the deterministic geometric signature of a spatially saturated universe.

References

Haimovich, T. (2026). " The Phantom Metric: Resolving Dark Matter Anomalies in Galactic Kinematics, Strong Lensing, and Galaxy Clusters via Topological Constraints and Topological Metric Deformation (Parameter-Free)" Zenodo Digital Repository. 10.5281/zenodo.22661163.